

Probabilistic Gradient Descent Bit Flipping (PGDBF) decoder for WiMAX LDPC codes.

<https://github.com/JurcaDavid10/WiMAX-LDPC-Decoder#>

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2. The Methodology

I utilized a parity-check matrix expanded by a factor of $Z=96$, resulting in a final dimension of 384×2304 . I addressed the "trapping set" problem by introducing two key improvements:

Energy Tolerance: I implemented a 0.5 energy tolerance to group bits with poor parity, ensuring I am not just flipping one bit at a time.

3. The Experimental Results

I performed a Bit Error Rate simulation to validate my decoder's performance under varying AWGN channel conditions.

Statistics across an SNR range of 4.0dB to 6.0dB.

This graph displays the "Waterfall" performance curve of my PGDBF decoder. As I increase the channel quality, we can observe a sharp decrease in the Bit Error Rate. This demonstrates that my decoder hits its operational threshold between 4.0dB and 5.2dB, where it successfully transitions from failing to correct high-noise signals to reliably recovering the transmitted data.

This Decoder is capable of achieving 100% success in moderate noise conditions (6.0dB)

